



3099704: AI for Digital Health

Multimodal Large Language Model (MLLM)

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Credit to TA.Pasith (Bass)



Outlines

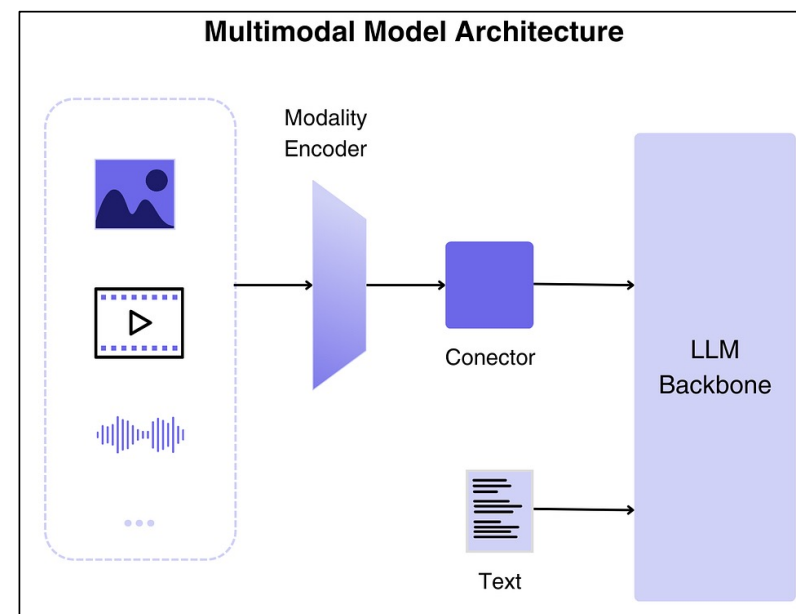
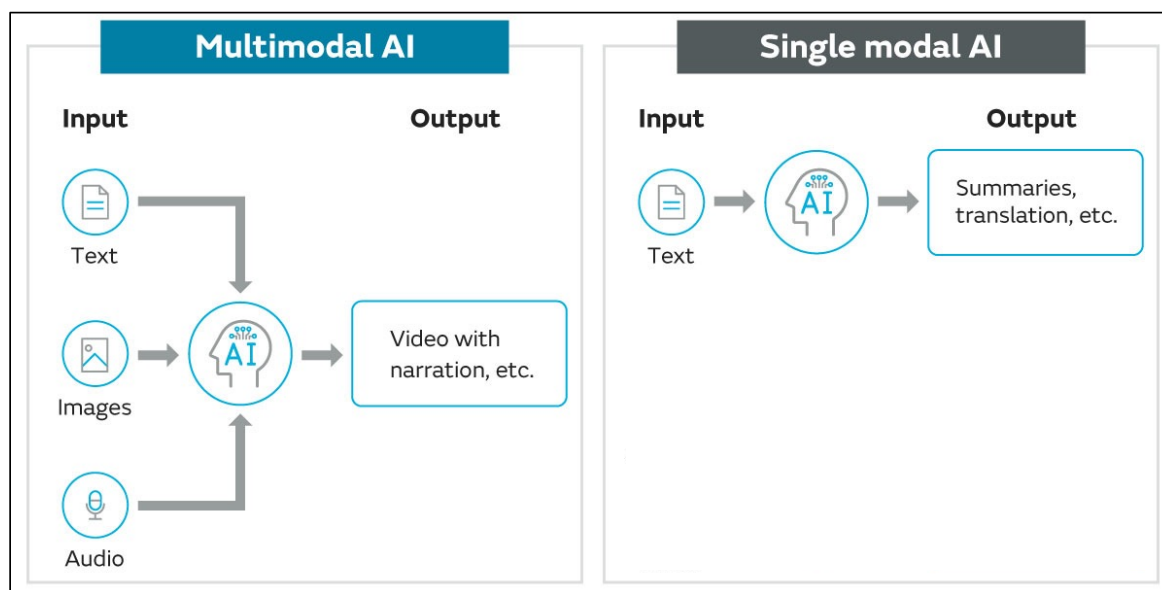
- MLLM Introduction
- MLLM Components
- Real World MLLM Applications
- MedGemma



Multimodal Large Language Model Introduction

What is MLLM?

A Multimodal Large Language Model (MLLM) is a foundation model that can **understand and reason over multiple input modalities**—such as text, images, audio, and video—and generate coherent language outputs.

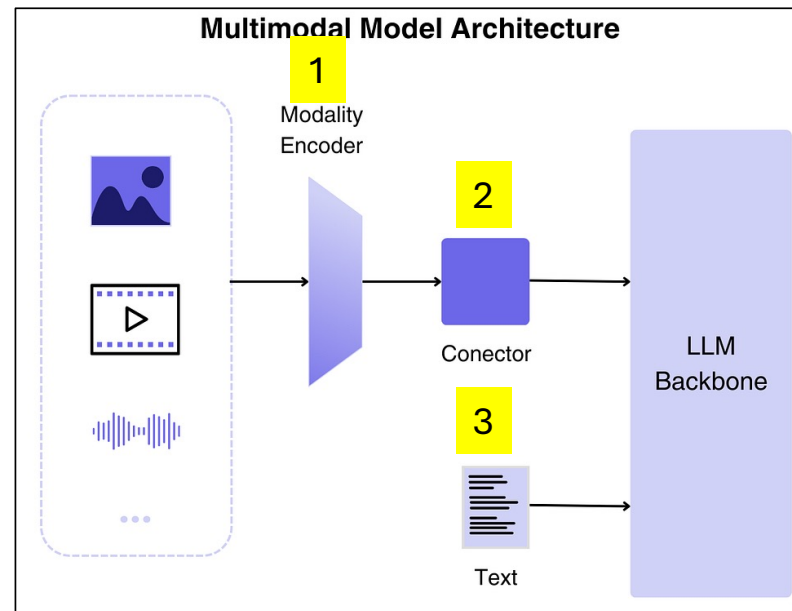




Multimodal Large Language Model Components

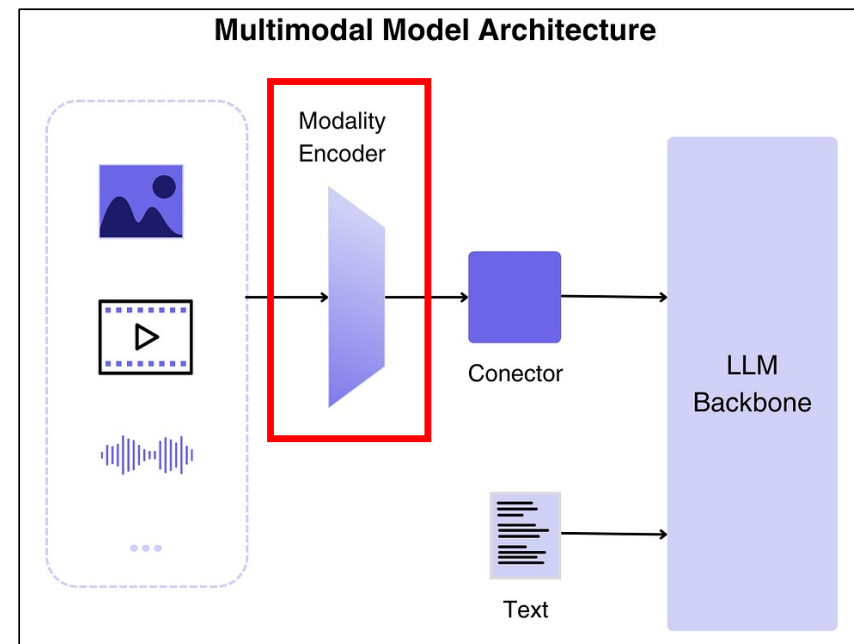
Core Components of MLLM

An MLLM architecture are mostly composed of (1) modality-specific encoders, (2) a connector layer, and (3) a large language model backbone that jointly enable perception and reasoning.



Core Components of MLLM

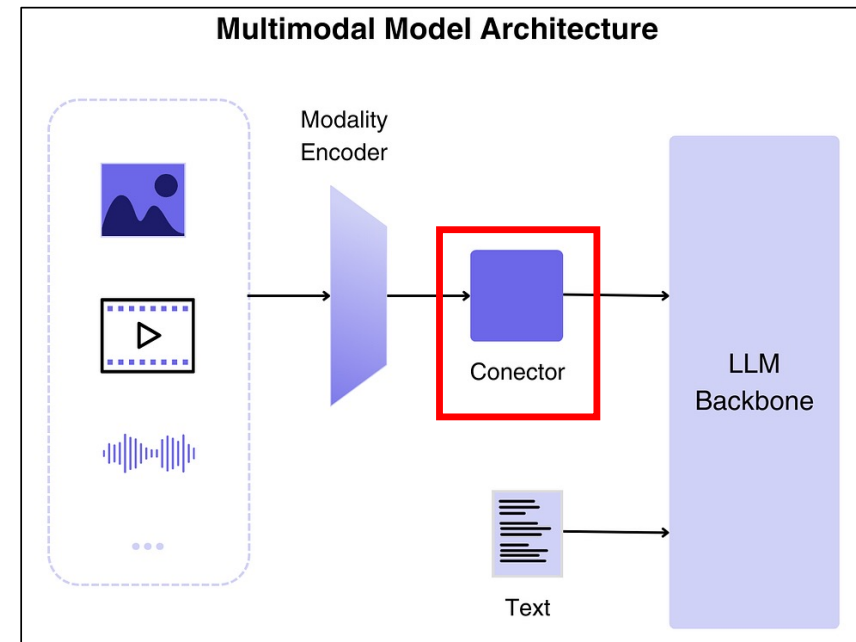
1. **Modality Encoders:** Convert raw inputs (images, video, audio, etc.) into vector representations. Each modality uses a specialized encoder
 - Image → Vision Encoder
 - Video → Vision + Temporal encoder
 - Audio → Audio Encoder



Core Components of MLLM

2. **Connector (Projection / Alignment Layer):** Bridges the gap between modality embeddings and LLM token space, which transforms non-text embeddings into a format the LLM can understand. Common connector implementations are:

- Linear projection
- MLP
- Cross-attention layers

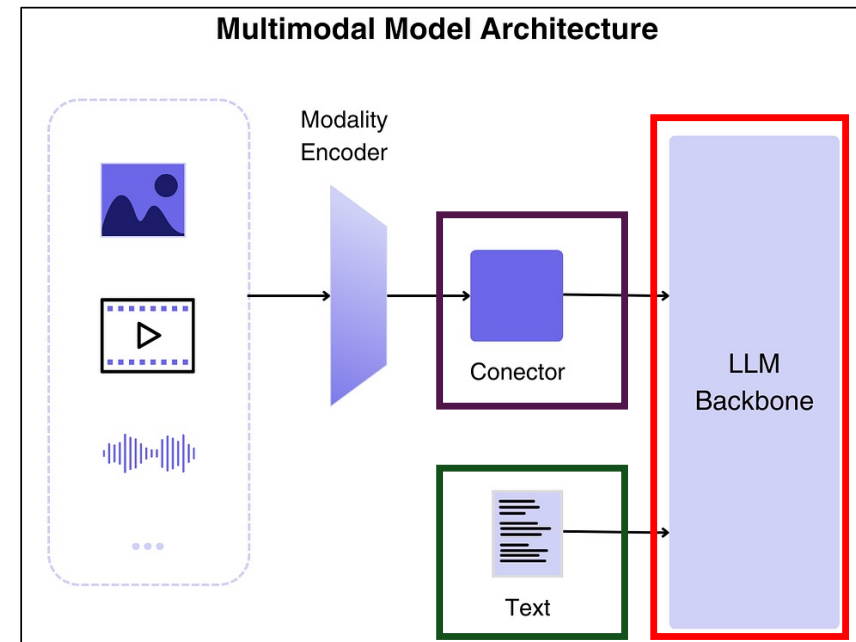


Core Components of MLLM

3. **LLM Backbone:** A pretrained Large Language Model to transform input from

- Text tokens (directly)
- Multimodal tokens (via the connector)

Then transform into textual explanation or any multimodal based on the architecture.

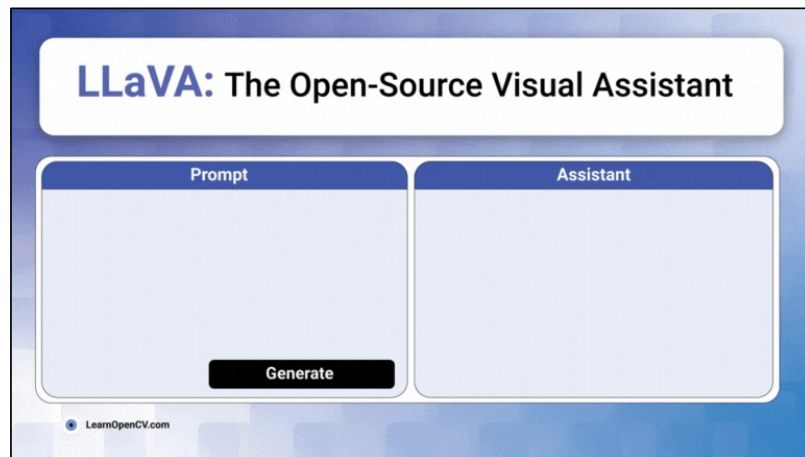


MLLM Applied to Images and Video: LLaVA

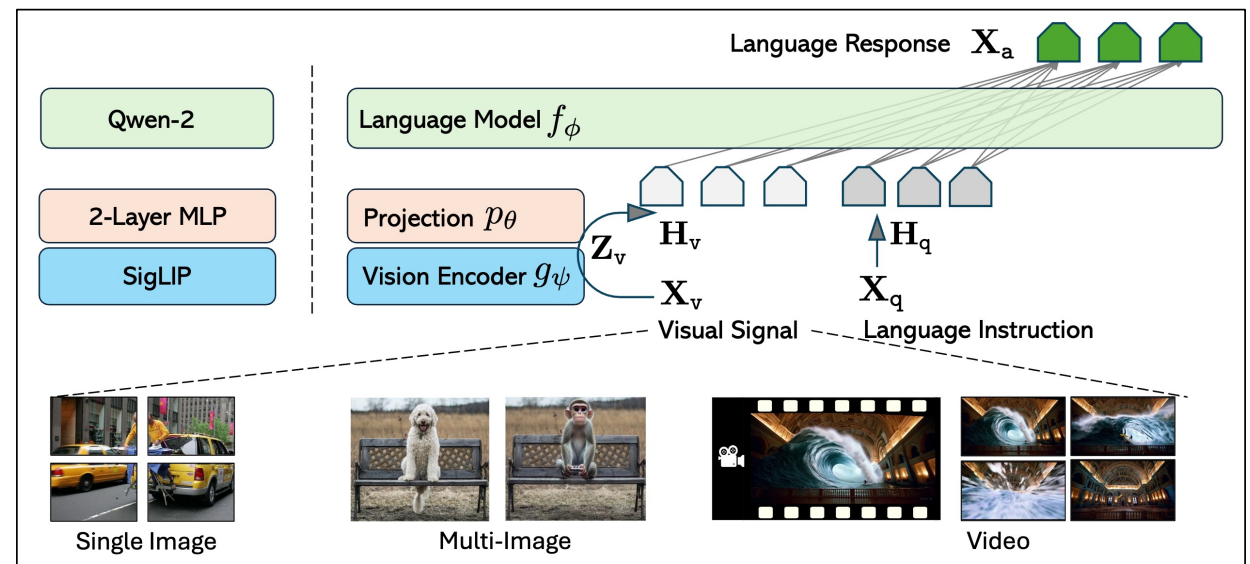
(Microsoft Research in NeurIPS-2023)

Large Language-and-Vision Assistant (LLaVA) is a multimodal large language model that enables instruction-following visual reasoning by combining a vision encoder with a language model design to answer natural language questions and follow visual instructions from images and video.

- **Modality Encoder:** CLIP, SigLIP Vision Encoder
- **Connector:** Multi Layer Perceptron
- **LLM Backbone:** LLaMA, Qwen-2



LLaVA: <https://arxiv.org/pdf/2304.08485>
LLaVA-NeXT: <https://arxiv.org/pdf/2407.07895>



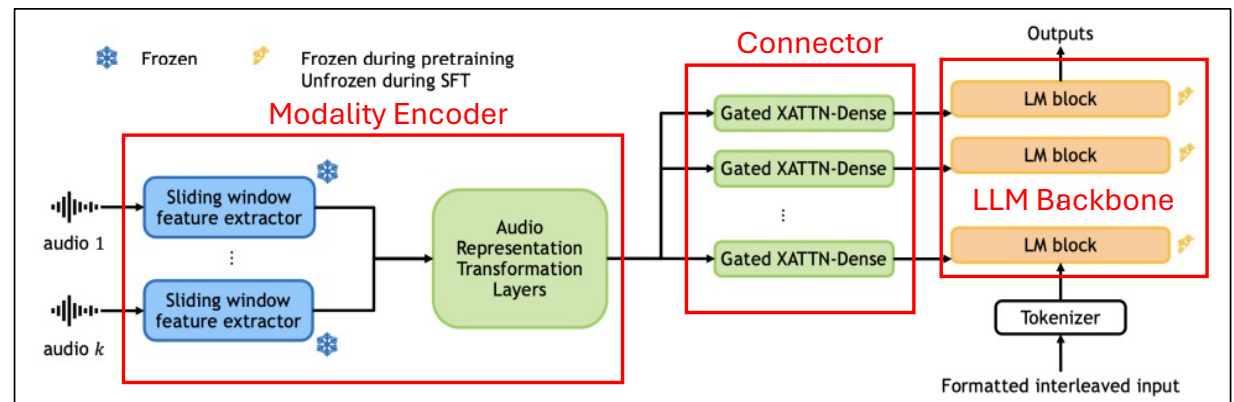
MLLM Applied to Audio: Audio Flamingo (Google DeepMind, 2023)

Audio Flamingo is a multimodal foundation model that generates and reasons about audio by combining audio representations with language modeling.

- **Modality Encoder:** ClapCap
- **Connector:** Gated cross-attention
- **LLM Backbone:** OPT-IML-MAX-1.3B

Audio Flamingo understands the order of sounds.
[Subjective description of the audio: a man speaks on a talk show, then the audience laugh and clap their hands.]
Task: *Audio Captioning*
Instruction: *Describe this sound.*
Ground Truth: *A man is speaking followed by applause.*
Baseline: *A crowd of people are speaking and applauding.*
Audio Flamingo: *A man is speaking and then the audience claps.*

Audio Flamingo understands occurrences in audio.
[Subjective description of the audio: three people talking in order, with a lot of noise in the recording.]
Task: *Audio Question Answering*
Instruction: *How many different voices are there?*
Ground Truth: *Three.*
Baseline: *One.*
Audio Flamingo: *Three.*

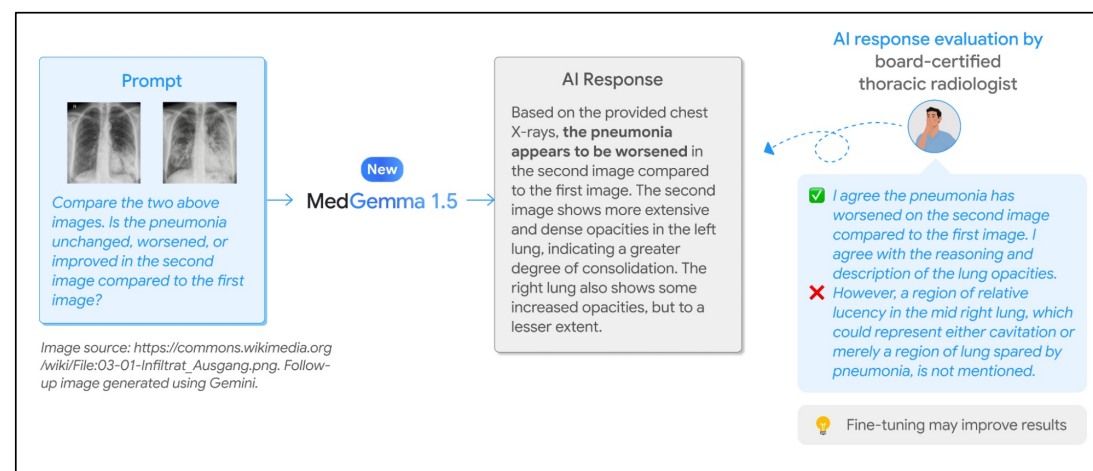
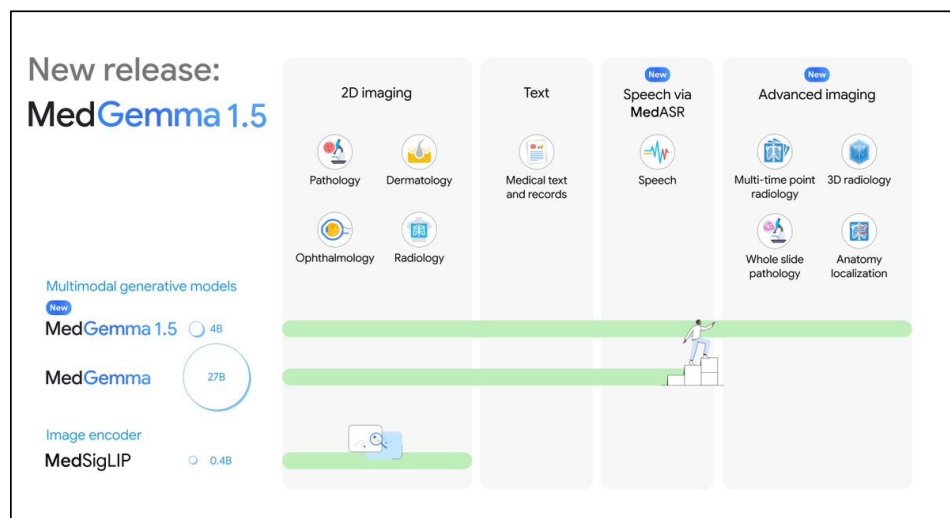




Real World Multimodal Large Language Model Applications

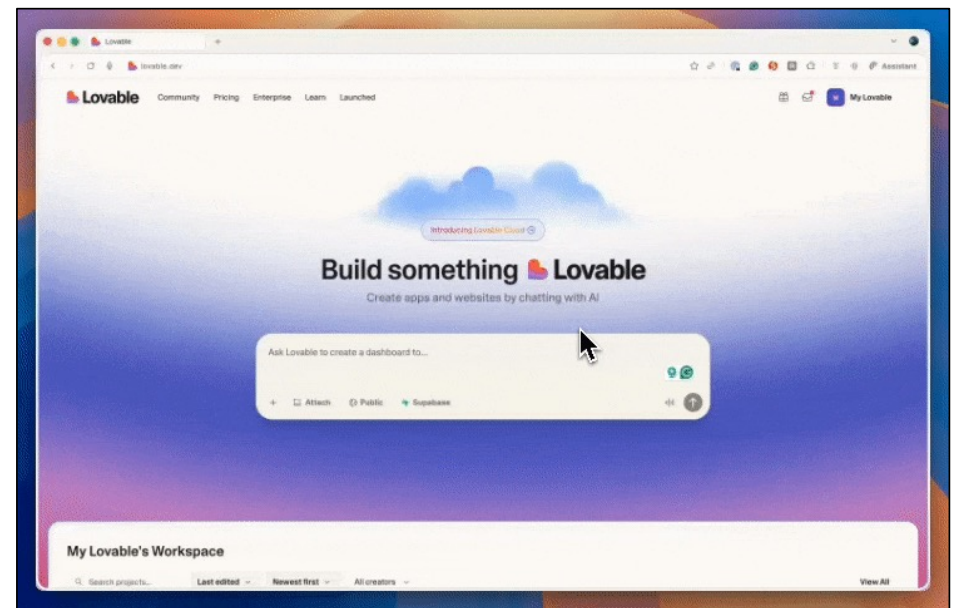
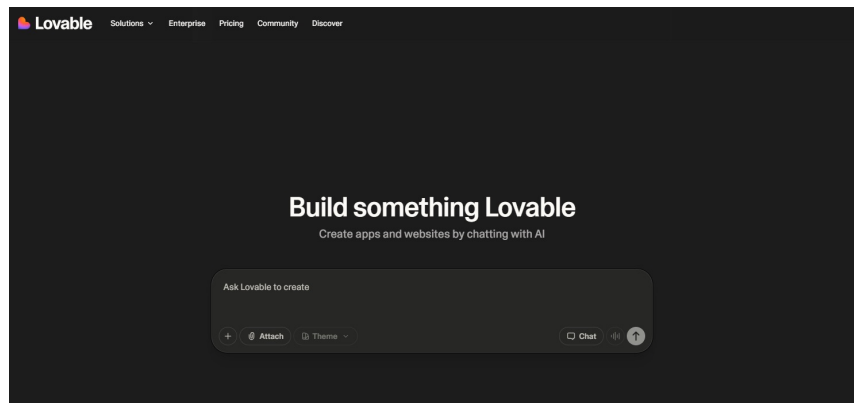
MedGemma — Healthcare Multimodal AI

MedGemma is a collection of multimodal medical AI models from Google/DeepMind designed for medical text and image understanding. It combines medical images (e.g., X-rays, CT, MRI) with textual clinical data to generate reports, assist reasoning, and extract structured insights from health records.



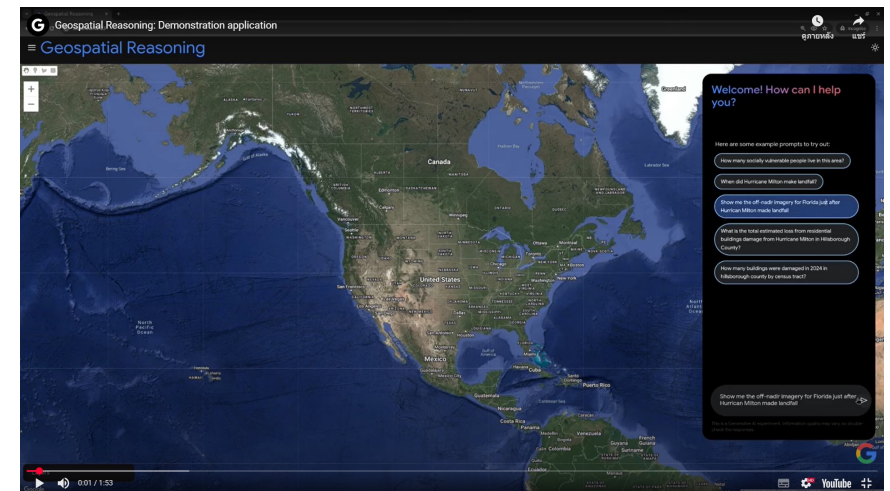
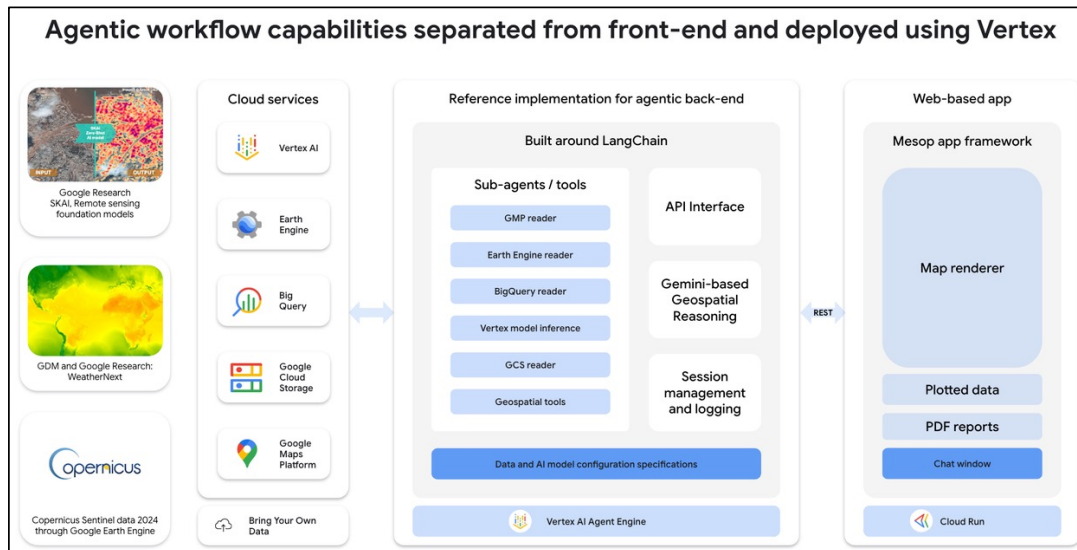
Lovable — AI-Powered App Development Platform

Lovable is a commercial AI application builder that uses powerful multimodal large language models to convert natural language prompts into real full-stack applications (frontend, backend, database, deployment).



Google Geospatial Reasoning

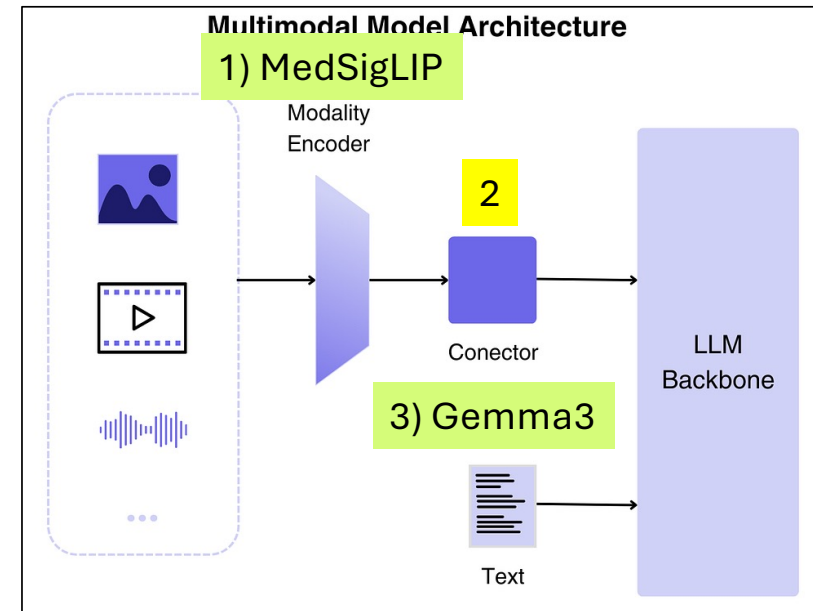
Google's Geospatial Reasoning initiatives use agentic multimodal AI + geospatial foundation models to analyze satellite imagery, maps, structured data, and text queries to generate insights about the Earth. These systems combine geospatial foundation models with multimodal reasoning to answer complex real-world questions about locations.





MedGemma

- What it is: A family of Gemma 3-based models optimized for medical text and image understanding
- History:
 - Developed by Google DeepMind / Google Research
 - Part of Google's Gemma open model family for domain-specific AI (Medical)
 - First announced: May 2025 (Google I/O timeframe)
 - Latest Version: MedGemma 1.5 released in January 2026
- Model Variants
 - 4B Multimodal
 - 27B Text-only
 - 27B Multimodal
- Multimodal Architecture: Uses MedSigLIP image (vision) encoder
- Pre-trained on de-identified medical images, including:
 - Chest X-rays
 - Dermatology & ophthalmology images
 - Histopathology slides





MedGemma (cont.)

Training Stages:

0. Vision finetuning: train [MedSigLIP](#) vision encoder.
1. Pretraining: SFT on [both medical domain and general domain](#) to prevent overfitting
2. Post-training:
 - Distillation: distil from larger LLM (e.g. medgemini, gemini)
 - [Reinforcement learning: Medical imaging data with paired text](#) was utilized in the RL stage of post-training.

MedGemma: Training Datasets

Table 1 | Overview of MedGemma training datasets.

Modality	Dataset	No. examples	Training stages	Description
Text-only	MedQA	9,275	Distill, RL	USMLE style exam questions
	MedMCQA	182,806	Distill, RL	Indian medical entrance exam questions
	AfriMed-QA	1,003	Distill, RL	Pan-African English multi-specialty QA
	MedExpQA	434	Distill, RL	Spanish medical residency exam questions
	PubMedQA	1,000	Distill, RL	Biomedical QA compiled from PubMed abstracts
	LiveQA	634	Distill, RL	Consumer health questions from United States National Library of Medicine (NLM)
	HealthSearchQA	3,375	Distill, RL	Common consumer medical questions from search engines
	Synthetic	200,000	Distill	Generated from large IT teacher
Radiology	SLAKE	450	Vision, PT, RL	Captions generated from QA pairs
	VQA-Rad	1,391	RL	Radiology image QA pairs
	MIMIC-CXR	231,483	Vision, PT, RL	Chest X-ray images & free-form reports
	Digital Knee X-ray	1,469	Vision, PT	Knee X-ray images & labels
	CT-US1	59,979	Vision, PT	2D CT slices & free-form reports
	MRI-US1	47,622	Vision, PT	2D MRI slices & free-form reports
Histopathology	Internal histopathology	32,550,599	Vision, PT, RL	Histopathology image patches, caption pairs
Dermatology	PAD-UFES-20	2,047	Vision, PT	Skin lesion images & labels
	Internal dermatology	51,049	Vision, PT, RL	Skin lesion images & labels
Ophthalmology	EyePACS	199,258	Vision, PT, RL	Fundus images & labels
General Medical	PMC	41,853	Vision, PT	Single panel medical images & caption pairs

Vision: Vision encoder enhancement, PT: Pretraining, Distill: Distillation, RL: Reinforcement learning.

MedGemma's Performance

Task and metric	Gemma 3 4B	MedGemma 4B	Gemma 3 27B	MedGemma 27B multimodal
Medical image classification				
MIMIC CXR** - macro F1 for top 5 conditions	81.2	88.9	71.7	90.0
CheXpert CXR - macro F1 for top 5 conditions	32.6	48.1	26.2	49.9
CXR14 - macro F1 for 3 conditions	32.0	50.1	31.4	45.3
PathMCQA* (histopathology, internal**) - Accuracy	37.1	69.8	42.2	71.6
US-DermMCQA* - Accuracy	52.5	71.8	66.9	71.7
EyePACS* (fundus, internal) - Accuracy	14.4	64.9	20.3	75.3
Visual question answering				
SLAKE (radiology) - Tokenized F1	40.2	72.3	42.5	70.3
VQA-RAD*** (radiology) - Tokenized F1	33.6	49.9	42.7	46.7
Knowledge and reasoning				
MedXpertQA (text + multimodal questions) - Accuracy	16.4	18.8	22.0	26.8

MedGemma's Performance (cont.)

The MedGemma models outperform their respective base Gemma models across all tested text-only health benchmarks.

Metric	Gemma 3 4B	MedGemma 4B	Gemma 3 27B	MedGemma 27B text-only	MedGemma 27B multimodal
MedQA (4-op)	50.7	64.4	74.9	89.8 (best-of-5) 87.7 (0-shot)	87.0 (best-of-5) 85.3 (0-shot)
MedMCQA	45.4	55.7	62.6	74.2	70.2
PubMedQA	68.4	73.4	73.4	76.8	77.2
MMLU Med	67.2	70.0	83.3	87.0	86.2
MedXpertQA (text only)	11.6	14.2	15.7	25.7	23.7
AfriMed-QA (25 question test set)	48.0	52.0	72.0	84.0	72.0

For all MedGemma 27B results, [test-time scaling](#) is used to improve performance.

Medical record evaluations

All models were evaluated on a question answer dataset from synthetic FHIR data to answer questions about patient records. MedGemma 27B multimodal's FHIR-specific training gives it significant improvement over other MedGemma and Gemma models.

Metric	Gemma 3 4B	MedGemma 4B	Gemma 3 27B	MedGemma 27B text-only	MedGemma 27B multimodal
EHRQA	70.9	67.6	84.2	86.3	90.5

+ Thank you
& any questions